INCH-POUND
MIL-DTL-49285/25B
31 August 2018
SUPERSEDING
MIL-DTL-49285/25A
1 March 2010

DETAIL SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, FIFTEEN INDIVIDUALLY SHIELDED PAIRS OF 20 (10 X 30) AWG

Inactive for new design after 16 June 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49285.

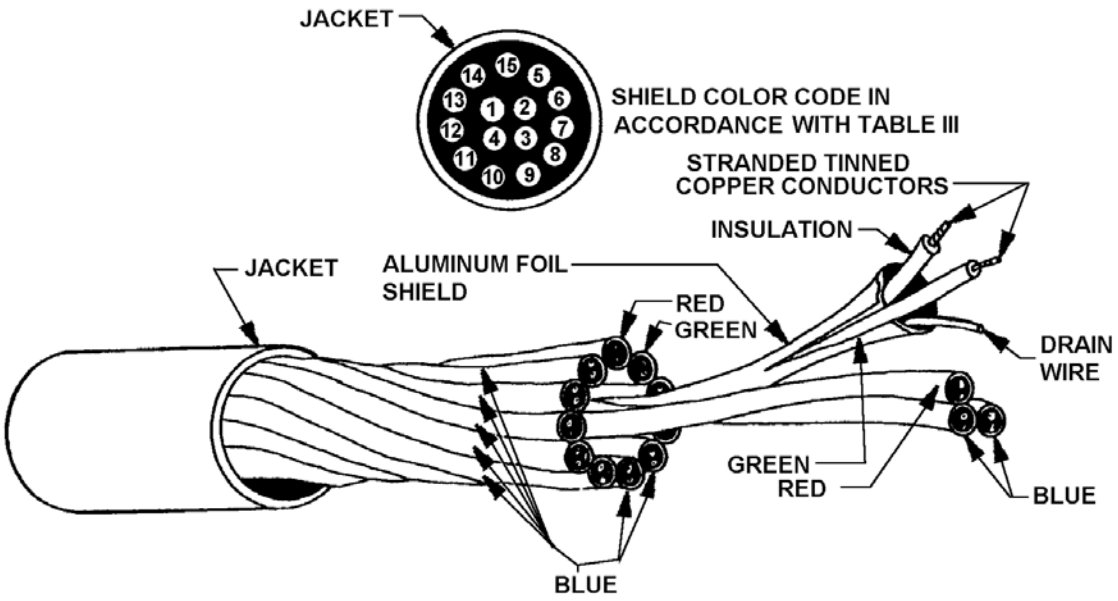


FIGURE 1. Cable illustration.

Engineering information: See table I.

Not recommended for shipboard or aerospace applications due to toxic smoke release in the event of a fire. PVC is not recommended in other applications (i.e. confined spaces, low ventilation) where there is the potential for human smoke exposure in a fire.

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TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors	30 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	55 pF/ft
Continuous working voltage	350 V rms maximum between conductors
Nominal capacitance between adjacent shields	150 pF/ft
Continuous working voltage between adjacent shields	50 volts maximum
Shield dc resistance	14 ohms/1,000 feet maximum

TABLE II. Description, physical.

Components	Construction details
Number of conductors	Fifteen individually shielded
Conductor type and wire size	Stranded tinned copper 20 AWG
Conductor stranding	10 X 30 AWG
Conductor insulation	Polypropylene
Conductor insulation thickness	0.013 inch nominal thickness
Drain wire type and size	Stranded tinned copper 22 AWG
Drain wire stranding	16 X 34 AWG
Jacket material	Polyvinyl chloride
Jacket thickness	0.050 inch
Finished cable diameter	0.688 inch maximum outside diameter
Cable style (UL)	2493
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay lengths	1.3 twists per foot $\pm$ 10 percent

Cabled pair orientation: The position of the red and green pair shield shall determine the rotation direction for pair location (see figure 1).

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TABLE III. Shield color code.

Pair numbers	Shield color
Center core	
1st pair	Red
2nd pair	Green
3rd pair	Blue – 1
4th pair	Blue – 2
Outer layer	
5th pair	Red
6th pair	Green
7th pair	Blue – 1
8th pair	Blue – 2
9th pair	Blue – 3
10th pair	Blue – 4
11th pair	Blue – 5
12th pair	Blue – 6
13th pair	Blue – 7
14th pair	Blue – 8
15th pair	Blue – 9

REQUIREMENTS:

Design and construction: (see table II):

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair, with the aluminum foil side inward. Insulation on the exterior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located beneath the shield with the insulated conductors. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Flammability. The cable shall pass UL 2556 flame test requirement.

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Shield integrity test.*

Frequency	Response
100 kHz	112 dB or more below reference
500 kHz	107 dB or more below reference
1 MHz	101 dB or more below reference
5 MHz	86 dB or more below reference
10 MHz	81 dB or more below reference
15 MHz	76 dB or more below reference
20 MHz	71 dB or more below reference
25 MHz	66 dB or more below reference
30 MHz	62 dB or more below reference

* Any two readings may deviate to a value of 0.9 times the stated limits.

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 40 dB below reference.

Crosstalk test limits. All values are minimums.

Frequency (kHz)	Near end crosstalk (dB)	Far end crosstalk (dB)
40	57	58
70	52	53
100	44	50
1,000	43	44

Insulation resistance between shields: 100 megohms-1,000 feet (minimum).

Durometer hardness. The cable jacket shall have a “Shore A” hardness of 82 $\pm$ 5.

Part or Identifying Number (PIN): The PIN shall be M49285/25.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-49285, this document references the following: UL 2556.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

Review activities:
Army – MI
Air Force – 71
DLA - CQ

(Project 6145-2018-041)

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